

Assessment – 3
Dry Run Test – Report
CSA1717 – Artificial Intelligence

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QUESTION 1 – Dry Run of A* Search Algorithm

1.1 Scenario Understanding

We are given a weighted graph with heuristic values for each node. The task is to perform a **step-by-step dry run of A*** from **A** $\rightarrow$ **G**, showing:

- Current node
- Open list
- Closed list
- $g(n)$, $h(n)$, $f(n)$
- Final optimal path and total cost

1.2 Heuristic Table

Node $h(n)$

A 7

B 6

C 4

D 3

E 2

G 0

1.3 A* Dry Run (Iteration-wise)*

Iteration 1

Current Node: A

Open List: A

Closed List: —

Node g(n) h(n) f(n)

A	0	7	7
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Expanding A

- B: $g=2, h=6 \rightarrow f=8$
- C: $g=4, h=4 \rightarrow f=8$

Updated Open List: B(8), C(8)

Closed List: A

Iteration 2

Current Node: B

Open List: C

Closed List: A, B

Node g(n) h(n) f(n)

B	2	6	8
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Expanding B

- C: new $g=5 \rightarrow f=9$ (worse than existing $f=8 \rightarrow$ ignore)
- D: $g=9 \rightarrow f=12$
- E: $g=4 \rightarrow f=6$

Updated Open List: E(6), C(8), D(12)

Iteration 3

Current Node: E

Open List: C, D

Closed List: A, B, E

Node g(n) h(n) f(n)

E 4 2 6

Expanding E

- G: $g=6 \rightarrow h=0 \rightarrow f=6$

Updated Open List: G(6), C(8), D(12)

Iteration 4

Current Node: G

Goal reached.

1.4 Optimal Path

$A \rightarrow B \rightarrow E \rightarrow G$

1.5 Total Path Cost

- $A \rightarrow B = 2$
- $B \rightarrow E = 2$
- $E \rightarrow G = 2$

Total Cost = $2 + 2 + 2 = 6$

FINAL ANSWER – QUESTION 1

Optimal Path: $A \rightarrow B \rightarrow E \rightarrow G$ ✓ Total Cost: 6

QUESTION 2 – Dry Run of Minimax with Alpha-Beta Pruning

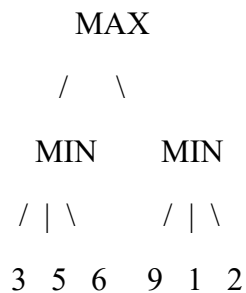
2.1 Scenario Understanding

We are given a game tree with MAX at the root and MIN at the next level. Leaf nodes contain utility values. We must perform Minimax with Alpha-Beta pruning left to right, showing:

- α (alpha)
- β (beta)
- Selected values
- Pruned nodes
- Final minimax value

2.2 Game Tree

Code



2.3 Dry Run (Step-by-Step)

Step 1 – MAX at Root

$$\alpha = -\infty \quad \beta = +\infty$$

Explore Left MIN subtree first.

Step 2 – MIN (Left Subtree)

Children: 3, 5, 6 $\alpha = -\infty \quad \beta = +\infty$

- Evaluate 3 $\rightarrow$ MIN picks 3 $\rightarrow \beta = 3$
- Evaluate 5 $\rightarrow$ MIN picks $\min(3,5)=3 \rightarrow \beta = 3$

- Evaluate 6 $\rightarrow$ MIN picks $\min(3,6)=3 \rightarrow \beta = 3$

Left MIN subtree value = 3

MAX updates: $\alpha = \max(-\infty, 3) = 3$

Step 3 – MIN (Right Subtree)

Children: 9, 1, 2 $\alpha = 3 \beta = +\infty$

- Evaluate 9 $\rightarrow \beta = 9$
 - No pruning ($9 > 3$)
- Evaluate 1 $\rightarrow \beta = \min(9,1) = 1$
 - Now $\beta = 1$
 - Check pruning: $\beta \leq \alpha \rightarrow 1 \leq 3 \rightarrow \text{TRUE}$

Node 2 is PRUNED

Right MIN subtree value = 1

Step 4 – MAX chooses best

MAX compares: $\max(3, 1) = 3$

FINAL ANSWER – QUESTION 2

Best Move for MAX: Left subtree

Final Minimax Value: 3

Pruned Nodes: 2

OVERALL SUMMARY

This assessment demonstrates:

- * efficiently finds the optimal path using cost + heuristic
- Minimax with Alpha-Beta pruning reduces computation by pruning unnecessary branches
- Both algorithms improve decision-making in AI systems under constraints